SPRDL1-24-R-0164 (JA-00859)
Previous J&A: JA-00517

JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION
SPRDL1-24-R-0164

1. **Contracting Agency:** Defense Logistics Agency - Land Warren (DLA-WRN).

2. **Description of Action:** DLA proposes to award a New Firm-Fixed-Price contract to Oshkosh Defense LLC, Commercial and Government Entity (CAGE) 75Q65, for National Stock Number (NSN) 2520-01-570-0617, Axle Assembly, Automotive, using Army Working Capital Funds.

3. **Description of Supplies/Services:** The NSN 2520-01-570-0617, part number 3831543, Axle Assembly, Automotive, is used on the Palletized Loading System (PLS) A1 and is a device suspended between and connecting opposite wheels. It supports the weight of a vehicle, and consists of the housing and axels, and may include the driving differential mechanism. The deliverable for this contract is for a base quantity of 3 each, with an option quantity of 3 each, making a total quantity of 6 each. The total maximum value of this contract is $[redacted]

Acquisition Method Code (AMC):
3 - Acquire, for the second or subsequent time, directly from the actual manufacturer.

Acquisition Method Suffix Code (AMSC):
R - The data or the rights to use the data needed to purchase this part from additional sources are not owned by the Government and it has been determined that it is uneconomical to purchase them.

4. **Authority Cited:** The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204(a)(1) as implemented by Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. **Reason for Authority Cited:** The PLS A1 axle assembly is a critical component of the entire PLS fleet. Axles play a crucial role in the vehicle's drivetrain assembly, transferring power generated by the engine from the transmission to the wheels. The wheels are mounted to the axle assembly, providing mobility to the vehicle. The PLS platform was designed and tested by Oshkosh, who owns the intellectual property for the truck. Oshkosh developed the system level requirements for the PLS and possesses details regarding the design intent of the axle. This axle was designed, configured, and integrated by Oshkosh for this specific vehicle.

The Government does not have the pertinent technical data to enable competitive re-procurement. The Government does not have rights to the TDP or specifications.

[redacted]

The axle is exclusively manufactured by Oshkosh, who holds all data rights related to these axle assemblies. Oshkosh possesses the specific technical expertise and proper facilities required for axle production. Furthermore, Oshkosh has a proven track record of building axle assemblies that enable PLS A1 vehicles to fulfill their mission. Oshkosh's axle assemblies have successfully operated in the field, meeting all mobility requirements outlined in the Army Technical Purchase Description. Unlike commercially produced axles, Oshkosh's designs are tailored specifically for use on the PLS platform. During the vehicle's design and initial production phases, Oshkosh and the Government conducted developmental and operational testing. These tests confirmed that Oshkosh could manufacture axles meeting the Government's performance and RAM requirements. If the Government were to purchase axle assemblies that have not undergone proper verification, mission failures and potential loss of mission critical functions to the vehicle could occur. Without this proper verification, it is likely that the axle would not fit or function properly, which could result in mobility failure, which would jeopardize mission capability and fleet readiness.

Since the Government does not own any of the technical data, the Government would have to reverse engineer the axle assembly to determine all aspects of the components. A reverse engineering effort would include work to clarify and define an axle with the same fit, form, and function as the original. Design requirements should be broken down and clarified without any ambiguity. Technical information such as understanding the parameters to include gear ratios, load ratings, housing dimensions, and mounting dimensions. Next, RAM testing would have to be completed. Testing would include accelerated life testing, cold and hot environments, salt spray, and maximum load operation testing. To neglect testing would be a high risk as premature failures of this item in the field could result in a mobility failure which could jeopardize the Army's mission in a potentially adverse operational environment. After the qualification, the TDP would have to be built so the part could be competitively sourced, and a source would have to be found.

The Ground Vehicle Systems Center (GVSC) Heavy Tactical Vehicle Sustainment Engineering Team has used historical data from past reverse engineering efforts to estimate that Reverse Engineering the PLS axle assembly would cost an estimated minimum of 54 months and $3,500,000.00 for:

- Research and Development (R&D), including part design, development, and prototype.
- RAM testing, design validation and vehicle integration testing; and
- TDP development, to procure competitively. A detailed breakdown of cost impacts is shown below.

Action	Cost
Part R&D (design, development, production)	$1,000,000.00
RAM Vehicle testing (5 vehicles at 5000 miles per vehicle)	$1,500,000.00
Develop TDP	$1,000,000.00
Total Development Time	**$3,500,000.00**

The Government would not be able to recover, through subsequent competition, the duplication of costs involved in either procuring the proprietary data or reverse engineering the item to create a competitive build to print technical data package within a reasonable time frame. At the current average usage rate of 9 units each year, a current unit cost of $[redacted] and assuming 25% savings achieved through competition, the $[redacted] in annual savings would take about [redacted] years to offset the estimated $3,500,000 cost of reverse engineering the item and qualifying another source.

6. Efforts to Obtain Competition:

a. <u>Effective competition</u>: Due to the reasons cited in section 5 above the Government is not pursuing competition for this item. Notices required by FAR 5.201 will be posted.

b. <u>Subcontracting competition</u>: Unless the contract is awarded to a small business, the solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in Subcontracting"; 52.219-16, "Liquidated Damages--Subcontracting Plan";" and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)," to maximize small business opportunities and utilization.

7. Actions to Increase Competition: There are no plans to increase competition for future procurements due to the reasons cited above.

8. Market Research: A Sources Sought Notice was posted to SAM.gov on 30 July, 2024, and was open for 15 days. No responses have been received to date.

The GVSC Tactical Vehicle Sustainment Engineer performed the following internet searches on 10 June 2024 to find suitable competition for the Oshkosh axle.

Keyword: axle assembly, heavy truck axle assembly, heavy duty axle assembly

Websites: https://www.cummins.com/product-finder?f%5B0%5D=product_type%3Aaxle
https://www.powertraininternationalweb.com/tag/axletech/
https://www.rockwellaxles.com/rockwell_axle_specs.htm

https://www.hendrickson-intl.com/search?st=axle

Findings: The axle product lines available from Cummins, AxleTech, Rockwell, and Hendrickson were reviewed for comparison. However, the government does not have any technical information for the Oshkosh axle because the data is proprietary, so further research could not be completed. Although several manufacturers make products that appear to be similar to the axle used on the PLS, the lack of a TDP or detailed drawings for the axle prohibit the engineer from knowing the exact specifications that must be met by a potential alternative. A wide range of component strength, maximum load, and hydraulic and air connections are available across the brands reviewed. However, the government does not have access to any of these characteristics for the provisioned axle, so a direct comparison is not possible. The current axle was designed by the OEM specifically to fit onto this vehicle, and integration/verification testing was conducted for this reason. Any other alternative would require integration and extensive testing to mitigate potential risks [redacted] Therefore, it can be concluded that this axle is custom for the application and should be identified as a sole-source item.

The Contracting Officer determined the Axle Assembly, Automotive, to be a commercial item based on procurement history. The prior award was made using FAR Part 12 procedures and the award is precedent setting in accordance with Defense Federal Acquisition Regulations (DFARS) 212.102(a)(ii)(A).

9. Interested Sources: To date no additional sources have expressed interest in this item.

10. Other Facts: Procurement History:

Procurement History:

(1) Contract: SPRDL1-23-C-0042
Award Date: 13 March 2023
Contractor: Oshkosh Defense, LLC (CAGE 75Q65)
Value: $1,220,200.05
Competitive Status: Limited Source
Authority: FAR 6.302-1, Only One Responsible Source

(2) Contract: SPRDL1-18-D-0030
Award Date: 27 February 2018
Contractor: Oshkosh Defense, LLC (CAGE 75Q65)
Value: $1,336,530.75
Competitive Status: Limited Source
Authority: FAR 6.302-1, Only One Responsible Source

(3) Contract: W56HZV-12-C-0420
Award Date: 14 September 2012
Contractor: Oshkosh Corporation (CAGE 45152)
Value: $747,690.80
Competitive Status: Limited Source
Authority: FAR 6.302-1, Only One Responsible Source

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [redacted]
Title: Engineering Branch Chief
Email: [redacted]@army.mil
Signature:

8/20/2024

X [redacted]

[redacted] [redacted]

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [redacted]
Title: Branch Chief – Procurement
Email: [redacted]@army.mil
Signature:

8/21/2024

X [redacted]

[redacted] [redacted]

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This determination will be based on the results of a detailed cost and/or price analysis.

The contractor may be required to submit other than certified cost and pricing data. This item has been determined to meet the FAR 2.101 definition for a commercial item and therefore certified cost and pricing data will not be required.

Name: Adam Reinbolt
Title: Contracting Officer
Email: adam.reinbolt@dla.mil
Signature:

8/21/2024

X Adam Reinbolt

Signed by: DLA

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Adam Reinbolt
Title: Contracting Officer
Email: adam.reinbolt@dla.mil
Signature:

8/21/2024

X Adam Reinbolt

Signed by: DLA